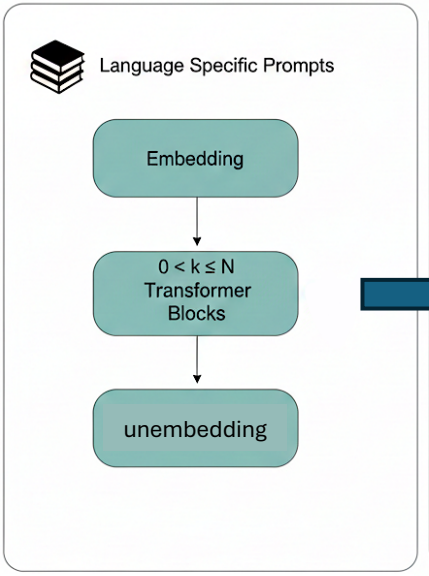
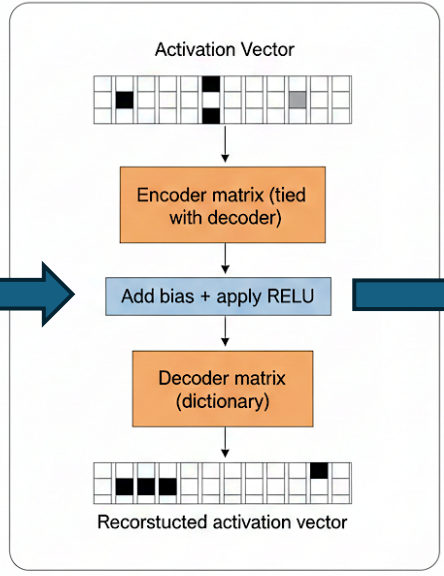
Language Model



Sparse Autoencoders



Language Neuron Dictionary

Neuron (ℓ,i)	Top Hindi tokens	Top Spanish tokens
ℓ=24,i=3102	मैं, है, के, और	yo, es, de, y
ℓ=26,i=1187	का, यह, नहीं	que, esta, no
ℓ=21,i=4420	समय, लोग, सरकार	tiempo, gente, gobierno
ℓ=28,i=0905	क्यों, कैसे	por qué, cómo
ℓ=19,i=2277	आज, कल, अभी	hoy, mañana, ahora
ℓ=23,i=5011	बहुत, अधिक, कम	muy, más, menos
ℓ=27,i=1349	करना, होना, जाना	hacer, ser, ir
ℓ=30,i=0788	क्या, कब, कहाँ	qué, cuándo, dónde

a. Sample activations from a language model

b. Learn a feature dictionary using an autoencoder that learns to represent activation vectors as a sparse linear combination of feature vectors.

c. Interpret the resulting language specific dictionary features.